

Tanmay Saxena

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Summary

AI/ML Engineer with 3+ years of software engineering experience and deep hands-on expertise building production AI systems, from multi-stage LLM pipelines and RAG architectures to real-time computer vision, speech processing, and 3D rendering. Skilled at integrating foundation models (Claude, GPT-4, Gemini) into end-to-end applications with sub-second latency, graceful degradation, and cost-controlled inference.

Technical Skills

AI/ML: Anthropic Claude API, OpenAI GPT-4o, Google Gemini 2.5, Deepgram Nova-2, ElevenLabs TTS
Computer Vision: MediaPipe Pose/Hand Landmarker, Multi-HMR + ANNY (163-bone mesh), 3D Gaussian Splatting
ML Infrastructure: RAG (Voyage AI embeddings + pgvector), DTW alignment, ONNX Runtime Web, Modal (GPU dispatch)
Languages: Python 3.11+, TypeScript 5.9, Go, JavaScript, Java, SQL
Frameworks: FastAPI, Next.js 16, React 19, React Native, Node.js, Django, PyTorch
Data & Infra: PostgreSQL (pgvector, async SQLAlchemy), Redis, AWS, GCP, Firebase, Docker, CI/CD
3D/Graphics: Three.js, WebGPU + WGSF shaders, custom GLSL shaders, Canvas 2D overlays

Experience

Q IT Technologies

June 2025 – Present

AI/ML Engineer (Lead)

Gainesville, FL

Directly — AI Dance Coaching Platform

- Built a real-time AI dance coaching platform achieving <150ms visual feedback latency — MediaPipe Pose Landmarker runs client-side at 30–60 FPS, transmitting only 165 floats/frame over WebSocket (privacy-by-design: video never leaves device).
- Designed a sliding-window Dynamic Time Warping alignment engine (dtadistance C backend) with weighted joint importance scoring, enabling tempo-agnostic pose comparison with per-frame similarity and 15-frame display smoothing.
- Engineered a bidirectional voice coaching loop — Deepgram Nova-2 WebSocket STT (echo guard + barge-in) → Claude Haiku intent classification → ElevenLabs Flash v2.5 TTS (~75ms); full round trip under 500ms with 528+ pre-cached phrases.
- Built a WebGPU-accelerated 3D Gaussian Splatting pipeline for photorealistic instructor overlay — GauHuman training on Modal serverless GPU, ONNX export (~12MB), custom WGSF shader with 3 visual styles and 3-tier graceful fallback.
- Implemented a RAG-enhanced coaching personalization system — Voyage AI voyage-3-lite (1024-dim) embeds coaching episodes into pgvector; cosine similarity retrieval ranked by 40% similarity + 60% effectiveness score, improving correction accuracy across sessions.
- Engineered predictive motion visualization with 19 movement archetypes classified in 3D ANNY joint space before 2D projection, plus velocity comet trails and kinematic chain sequencing badges; implemented real-time RNN beat detection via madmom with hierarchical choreography segmentation.

Shorten — AI Video Intelligence Platform

- Architected a 6-stage LLM pipeline (Gemini 2.5 Pro) generating 80–120 page enterprise reports from 50–100 aggregated video signals, consuming ~40K–60K tokens per report at \$0.40–0.60/brand.
- Built multimodal video analysis integrating Deepgram transcription, Ollama local vision models, and Groq for signal synthesis — extracting sentiment, sarcasm, product claims, and competitor mentions from raw video.
- Designed a graceful AI fallback chain (Gemini multimodal → Deepgram + Ollama + Groq → GPT-4o); implemented semantic relevance verification using OpenAI embeddings to filter multi-source evidence before surfacing signals.

FavorIt — Peer-to-Peer Marketplace App

- Engineered a Generative AI receipt processing feature integrating Vision OCR and GPT-4, cutting settlement time by over 80%; implemented bounding box-based POI detection for real-time location discovery.

Tata Consultancy Services (TCS)

Jan 2021 – May 2023

Systems Engineer (Developer – General Motors)

Bengaluru, India

- Automated data processing with Django-React app, reducing manual work by 87.5%; optimized 1TB+ PostgreSQL dataset with indexing and partitioning, improving query performance by 60%.

Education

University of Florida

Aug 2023 – May 2025

M.S. in Computer Science (GPA: 3.90/4.00)

Gainesville, FL

SRM Institute of Science and Technology

Jun 2016 – May 2020

B.Tech in Electronics and Communications Engineering (GPA: 8.58/10.00)

Chennai, India